Prompt Log for Software Design Document

Below is a chronological list of user prompts and inputs that were used to generate the Software Design Document (SDD) for the Smart Maintenance Application project. These guided the analysis, structure, and generation of architecture, UML diagrams, and detailed design documentation.

Project Overview + Architecture

- • Hello, I am trying to create a design document for a programming project that I have been working on all semester. I have attached a template that includes instructions and examples. What information would you need from me to create one for my project?
- • For my project, myself and 5 others have worked in partnership with the NSA to develop a smart maintenance application. We use arduinos to simulate data from a train's Air Compression Unit. It then sends that information to a raspberry pi via bluetooth. The raspberry pie then batches the readings and sends the info to our database which is hosted in DynamoDB using AWS. The data is run through an isolation forest model to test for anomalies in the components. The trained model is also stored on AWS in an S3 bucket. On the Machine Learning side we also have a model which takes this data and using a batching process, determines an optimal schedule for maintenance on different trains. On the backend we use python running a Flask API in order to connect to databases, perform authentication, and set up routes for many different needs. On the front end we present this model as a website using react. The website has an option for users to log in, signup, logout, view components of trains if they are logged in on their dashboard, and view the schedule on the schedule page. Each component has an expected days until failure property that informs when they should be scheduled for maintenance
- • 1. This is currently functioning as a prototype for the NSA and is a school project so it is unlikely that it will be used for others as of now. If I had to guess I would say it could be used by maintenance engineers looking to optimize their repair schedules.
- • 2. I can give you the code for the modules you need, we have split it up into github repos, including an embedded systems repo which has the raspberry pi and arduino code and is mostly in C, a machine learning repo which is made for training models and is in python, and the main repo which includes front end (react) and backend (flask/python)

Repository Uploads (Architecture & Design Structure)

- • How should I download the zips from github?
- • https://github.com/ITSC-4155-002/Main/tree/staging#
- • https://github.com/ITSC-4155-002/MachineLearning
- • https://github.com/ITSC-4155-002/EmbeddedSystems

UML Diagram Requests

- • Ready
- • Can you complete it all without me saying ready for each step?
- • Can you insert them for me?
- • Can you give me a PDF?

Final Touches & Deliverables

- • Where do the UML designs I've downloaded from you go in the design document?
- • can you give me all of the prompts I asked that were used in the creation of the doc

Other AI Contributions

GitHub

- GitHub Copilot was used to generate reports and changelogs for releases and merge reviews.

ChatGPT

- ChatGPT was used to create meeting itineraries.
- ChatGPT was also used to help with README.md formatting.
- ChatGPT was used to create images used on the website.